

14.3 Multiple correlation

The population squared multiple correlation

$$\rho_{Y \cdot X}^2 = \frac{\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\sigma_Y^2}$$

is the largest squared correlation between Y and a linear combination of $\mathbf{X}$ (Proposition 2.5.3). Section 9.5 studied its sample counterpart R^2 for a fixed design (Theorem 9.5.1, Theorem 9.5.2). With random rows, R^2 estimates $\rho_{Y \cdot X}^2$. This section finds its exact distribution under multivariate normality, and uses it to test $\rho_{Y \cdot X}^2 = 0$, to build an interval and to measure the bias.

Throughout, $(\mathbf{X}_i^\top, Y_i)$, $i = 1, \dots, n$, are independent draws from a $(k+1)$ -variate normal distribution with $\boldsymbol{\Sigma}_{XX}$ positive definite and $\sigma_Y^2 > 0$, and $n \geq k+2$. By Eq. 3.5.2, $Y_i = \alpha + \boldsymbol{\beta}^\top \mathbf{X}_i + e_i$ with independent $e_i \sim N(0, \sigma^2)$ independent of the regressors, where

$$\sigma^2 = \sigma_Y^2(1 - \rho_{Y \cdot X}^2), \quad \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta} = \sigma_Y^2 \rho_{Y \cdot X}^2, \quad \text{so} \quad \frac{\boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}}{\sigma^2} = \frac{\rho_{Y \cdot X}^2}{1 - \rho_{Y \cdot X}^2}. \quad (14.3.1)$$

The ratio on the right is the population *signal-to-noise ratio*. We write ρ^2 for $\rho_{Y \cdot X}^2$ when no confusion can arise.

14.3.1 The sample multiple correlation as an estimator

Let $\mathbf{S}_{XX}$, $\mathbf{s}_{XY}$ and s_Y^2 be the sample covariances, with any common divisor. The fit with an intercept has $\text{SSR} = \hat{\boldsymbol{\beta}}^\top \mathbf{W} \hat{\boldsymbol{\beta}}$, where $\mathbf{W}$ is the centred cross-product matrix of the regressors and $\hat{\boldsymbol{\beta}} = \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(Y_i - \bar{Y})$ (Theorem 6.6.2 with $\mathbf{X}_2 = \mathbf{1}$). Dividing by SST gives

$$R^2 = \frac{\mathbf{s}_{XY}^\top \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}}{s_Y^2}, \quad (14.3.2)$$

the population formula with sample moments in place of population moments. Under normality it is more than a plug-in.

Proposition 14.3.1 — Maximum likelihood with normal rows

Suppose the sample covariance matrix of the rows is nonsingular. The maximum likelihood estimates of $(\alpha, \boldsymbol{\beta}, \sigma^2)$ are the least squares estimates and SSE/n , and the maximum likelihood estimate of $\rho_{Y \cdot X}^2$ is R^2 .

Proof. The joint density of a row factorizes as $f(\mathbf{x}; \boldsymbol{\mu}_X, \boldsymbol{\Sigma}_{XX}) f(y | \mathbf{x}; \alpha, \boldsymbol{\beta}, \sigma^2)$, a normal density for the regressors times the conditional normal density of Theorem 3.5.2. The map from $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ to $(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_{XX}, \alpha, \boldsymbol{\beta}, \sigma^2)$ is one-to-one onto the set where $\boldsymbol{\Sigma}_{XX}$ is positive definite and $\sigma^2 > 0$, with inverse $\boldsymbol{\sigma}_{XY} = \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}$, $\sigma_Y^2 = \sigma^2 + \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}$, $\mu_Y = \alpha + \boldsymbol{\beta}^\top \boldsymbol{\mu}_X$. So the log-likelihood is a sum of two terms with separate parameters, and each can be maximized on its own. The conditional term is the likelihood of a normal linear model

with model matrix $[\mathbf{1}, \mathbf{X}_1]$, maximized by least squares and SSE/n (Theorem 7.3.1).

In the marginal term the quadratic form is minimized over $\boldsymbol{\mu}_X$ at $\bar{\mathbf{x}}$, leaving $-\frac{n}{2} [\log \det \boldsymbol{\Sigma}_{XX} + \text{tr}(\boldsymbol{\Sigma}_{XX}^{-1} \mathbf{S})]$ with $\mathbf{S} = \mathbf{W}/n$ positive definite. If $\lambda_1, \dots, \lambda_k > 0$ are the eigenvalues of $\mathbf{S}^{1/2} \boldsymbol{\Sigma}_{XX}^{-1} \mathbf{S}^{1/2}$, then

$$\log \det \boldsymbol{\Sigma}_{XX} + \text{tr}(\boldsymbol{\Sigma}_{XX}^{-1} \mathbf{S}) - \log \det \mathbf{S} - k = \sum_{j=1}^k (\lambda_j - \log \lambda_j - 1) \geq 0,$$

with equality iff every $\lambda_j = 1$, that is, iff $\boldsymbol{\Sigma}_{XX} = \mathbf{S}$. By invariance of maximum likelihood under reparameterization, the estimate of $\rho^2 = \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta} / (\sigma^2 + \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta})$ is $\hat{\boldsymbol{\beta}}^\top \mathbf{S} \hat{\boldsymbol{\beta}} / (\text{SSE}/n + \hat{\boldsymbol{\beta}}^\top \mathbf{S} \hat{\boldsymbol{\beta}}) = \text{SSR} / (\text{SSE} + \text{SSR}) = R^2$. ■

The rest of the section concerns the sampling distribution of this estimator.

14.3.2 The distribution of R-squared

Theorem 14.3.2 — Distribution of R^2 with normal rows

Under the assumptions above, let $F = \frac{n-k-1}{k} \frac{R^2}{1-R^2}$ and $\lambda = \rho^2/(1-\rho^2)$.

- (a) Given the regressors, $F \sim F(k, n-k-1, \gamma)$ with $\gamma = \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} / \sigma^2$, where $\mathbf{W}$ is the centred cross-product matrix of the regressors.
- (b) If $\rho^2 = 0$, then $F \sim F(k, n-k-1)$ and $R^2 \sim \text{Beta}(\frac{k}{2}, \frac{n-k-1}{2})$, independently of the regressors. The overall F test of the regression is an exact size- α test of $\rho^2 = 0$, which under joint normality is the hypothesis that Y is independent of $\mathbf{X}$.
- (c) If $\rho^2 > 0$, then $\gamma = \lambda V$ with $V \sim \chi^2(n-1)$, and for $0 < c < 1$

$$\Pr(R^2 \leq c) = \mathbb{E} \left[\Pr \left\{ F(k, n-k-1, \lambda V) \leq \frac{(n-k-1)c}{k(1-c)} \right\} \right]. \quad (14.3.3)$$

- (d) For each $c \in (0, 1)$, $\Pr(R^2 > c)$ is a strictly increasing function of $\rho^2 \in [0, 1)$.

Proof.

- (a) The centred regressor matrix has rank k with probability one, because $\mathbf{W} \sim W_k(n-1, \boldsymbol{\Sigma}_{XX})$ (Lemma 14.2.3) is nonsingular when $n-1 \geq k$ (the nonsingularity step in the proof of Lemma 14.2.4 needs only $m \geq k$). Given the regressors, the model is a normal linear model with $r = k+1$ and mean $\alpha \mathbf{1} + \mathbf{X}_1 \boldsymbol{\beta}$. By Theorem 9.5.2(a), $F \sim F(k, n-k-1, \gamma)$ with $\gamma = \|(\mathbf{M} - \mathbf{P}_1)(\alpha \mathbf{1} + \mathbf{X}_1 \boldsymbol{\beta})\|^2 / \sigma^2$. Since $(\mathbf{M} - \mathbf{P}_1)\mathbf{1} = \mathbf{0}$ and $(\mathbf{M} - \mathbf{P}_1)\mathbf{X}_1 \boldsymbol{\beta} = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1 \boldsymbol{\beta}$, this is $\boldsymbol{\beta}^\top \mathbf{X}_1^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{X}_1 \boldsymbol{\beta} / \sigma^2 = \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} / \sigma^2$.
- (b) $\rho^2 = 0$ means $\boldsymbol{\sigma}_{XY} = \mathbf{0}$, so $\boldsymbol{\beta} = \mathbf{0}$ and $\gamma = 0$ for every design. By Lemma 14.1.1, $F \sim F(k, n-k-1)$ independently of the regressors, and so is $R^2 = kF/(kF + n - k - 1)$, which is $\text{Beta}(k/2, (n-k-1)/2)$ by Theorem 9.5.2(b). For jointly normal variables, zero covariance is independence (Theorem 3.4.1).
- (c) By the proof of Lemma 14.2.3, $\mathbf{W} = (\mathbf{K} \mathbf{X}_1)^\top (\mathbf{K} \mathbf{X}_1)$, where the $n-1$ rows of $\mathbf{K} \mathbf{X}_1$

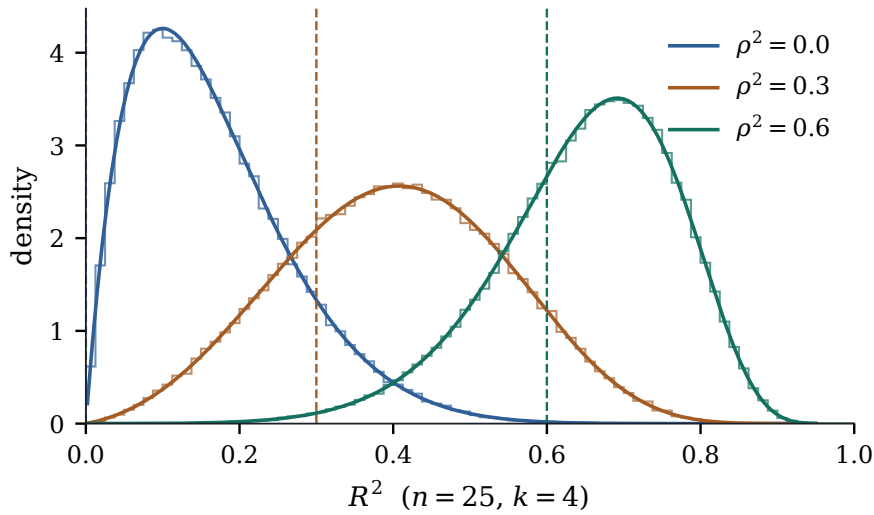


Figure 14.3.1: The law of R^2 with normal rows, $n = 25$, $k = 4$, for $\rho^2 = 0, 0.3$ and 0.6 (dashed). Step outlines: simulation. Curves: Eq. 14.3.3.

are independent $N_k(\mathbf{0}, \Sigma_{XX})$. So $\mathbf{KX}_1\boldsymbol{\beta}$ has independent $N(0, \boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta})$ entries, and $\boldsymbol{\beta}^\top \mathbf{W}\boldsymbol{\beta} = \|\mathbf{KX}_1\boldsymbol{\beta}\|^2 = (\boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta}) V$ with $V \sim \chi^2(n-1)$. By Eq. 14.3.1, $\gamma = \lambda V$. Since $R^2 \leq c$ iff $F \leq (n-k-1)c/(k(1-c))$, taking expectations of the conditional probability from (a) gives Eq. 14.3.3.

- (d) Fix $v > 0$ and $f > 0$. By Theorem 4.1.4, $\Pr\{F(k, n-k-1, \lambda v) > f\}$ is strictly increasing in λ , and $\lambda = \rho^2/(1-\rho^2)$ is strictly increasing in ρ^2 . If $\rho_1^2 < \rho_2^2$, the conditional tail probabilities therefore differ by a positive amount for every value of $V > 0$, and so do their expectations.

■

Part (c) is Fisher's (1928) distribution of the multiple correlation coefficient, reached here without any integration. Given the design, R^2 is a transformed noncentral F variable, and the design enters only through the single number $\boldsymbol{\beta}^\top \mathbf{W}\boldsymbol{\beta}$, which is a scaled chi-squared variable because the regressors are normal. Figure 14.3.1 shows the resulting densities for $n = 25$ and $k = 4$, computed from Eq. 14.3.3 by averaging over a grid of quantiles of V .

Example. (How biased is R-squared?)

With $n = 25$, $k = 4$ and 100000 simulated samples, the averages of R^2 are 0.1669, 0.4026 and 0.6500 for $\rho^2 = 0, 0.3$ and 0.6 , against exact values 0.1667, 0.4026 and 0.6498. The exact values come from the formula $E(R^2) = 1 - \frac{n-k-1}{n-1}(1-\rho^2) {}_2F_1(1, 1; \frac{n+1}{2}; \rho^2)$, which follows from Fisher's (1928) distribution of R^2 (see Olkin and Pratt 1958), where ${}_2F_1$ is the hypergeometric function; it can also be computed from Eq. 14.3.3. The bias shrinks as ρ^2 grows. The adjusted R^2 of Eq. 9.5.1 averages 0.0003, 0.2831 and 0.5800: unbiased under the null (Proposition 9.5.3(e)), but biased *downwards* when $\rho^2 > 0$. Neither is unbiased everywhere. The standard deviations of R^2 are 0.104, 0.146 and 0.118: with 25 cases, a single R^2 says little about ρ^2 .

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats
```

```
def r2_cdf(r, rho2, n, k, m=4000):
    """P(R^2 <= r) for iid normal rows: average the conditional noncentral F over V.
    """
    df2 = n - k - 1
    f = df2 * r / (k * (1 - r))
    if rho2 == 0:
        return stats.f.cdf(f, k, df2)
    V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)      # equal-probability
    nodes
    return np.mean(stats.ncf.cdf(f, k, df2, rho2 / (1 - rho2) * V))
```

[Run this code online](#)

Part (d) says that the overall F test is unbiased and has power increasing in ρ^2 , as a test of $\rho^2 = 0$ against random-regressor alternatives. For $n = 25$ and $k = 4$ it rejects when $R^2 > 0.364$, and its power is 0.182, 0.377, 0.599 and 0.922 at $\rho^2 = 0.1, 0.2, 0.3$ and 0.5. A population in which the regressors account for a fifth of the variance of the response is detected less than half the time.

14.3.3 An interval for the squared multiple correlation

Part (d) also makes an exact confidence interval possible. For observed $R^2 = r$, let ρ_L^2 be the value at which $\Pr(R^2 \leq r) = 0.975$ and ρ_U^2 the value at which it equals 0.025 (with $\rho_L^2 = 0$ if $\Pr(R^2 \leq r) < 0.975$ already at $\rho^2 = 0$). Because the probability decreases in ρ^2 , $[\rho_L^2, \rho_U^2]$ covers the true ρ^2 exactly when R^2 lies between the 2.5% and 97.5% points of its law at that ρ^2 , which has probability 0.95. This is the usual inversion of a family of tests (Proposition 12.1.2).

Example. (Violent crime in the states)

For the 50 states in 2009, regress the violent crime rate on the high-school graduation rate, the poverty rate, the percentage of single-parent households, the percentage of white residents and the urbanization rate. Then $R^2 = 0.4607$, the adjusted value is 0.3994, and $F = 7.52$ on 5 and 44 degrees of freedom, with p -value 3.6×10^{-5} . Read with random regressors, the 95% interval for ρ^2 is (0.175, 0.610). The interval is wide: five regressors and fifty cases leave much room for chance. It also treats the states as a random sample, which they are not; it describes hypothetical replications of a process generating rows like these, and here, as in Example (Poverty and graduation), no more can be claimed.

```
data = sm.datasets.statecrime.load_pandas().data.drop(index="District of Columbia")
regs = ["hs_grad", "poverty", "single", "white", "urban"]
y = data["violent"].to_numpy()
X = np.column_stack([np.ones(len(y))] + [data[c].to_numpy() for c in regs])
fit = sm.OLS(y, X).fit()
n_s, k_s = len(y), len(regs)
R2_obs = fit.rsquared
print(f"R^2 = {R2_obs:.4f}, adjusted {fit.rsquared_adj:.4f}, F = {fit.fvalue:.2f}, "
      f"p = {fit.f_pvalue:.2g}")
```

```
# 95% interval for rho^2: the values at which the observed R^2 is not extreme
def limit(prob):
    g = lambda rho2: r2_cdf(R2_obs, rho2, n_s, k_s) - prob
    return 0.0 if g(1e-12) < 0 else optimize.brentq(g, 1e-12, 0.999)

lower, upper = limit(0.975), limit(0.025)
print(f"95% interval for rho^2: ({lower:.3f}, {upper:.3f})")
```

[Run this code online](#)

Remark (Fixed or random?). Under a fixed design the expectation of R^2 depends on the chosen regressor values (Exercise (B1)). With random rows R^2 estimates $\rho_{Y \cdot X}^2$, which still depends on the regressor distribution: a narrower range gives a smaller ρ^2 . The null law of the F test is the same in both readings.

14.3.4 Exercises

14.3.4.1 A. Check your understanding

Exercise A1.

A regression on $k = 3$ random regressors from $n = 40$ cases has $R^2 = 0.25$. Compute the overall F statistic and its degrees of freedom. Under the null hypothesis $\rho^2 = 0$, what are the mean and the distribution of R^2 ?

14.3.4.2 B. Practice

Exercise B1.

Let $\mathbf{R}$ be the sample correlation matrix of $(Y, X_1, \dots, X_k)$, assumed nonsingular, and let r^{YY} be the $(1, 1)$ entry of $\mathbf{R}^{-1}$. Show that $R^2 = 1 - 1/r^{YY}$, and that $1 - R^2 = \det \mathbf{R} / \det \mathbf{R}_{XX}$, where $\mathbf{R}_{XX}$ is the correlation matrix of the regressors.

Solution. Eq. 14.3.2 is unchanged when every variable is standardized, so $R^2 = \mathbf{r}_{XY}^\top \mathbf{R}_{XX}^{-1} \mathbf{r}_{XY}$. By Theorem 1.6.4, the leading entry of $\mathbf{R}^{-1}$ is the inverse of the Schur complement $1 - \mathbf{r}_{XY}^\top \mathbf{R}_{XX}^{-1} \mathbf{r}_{XY} = 1 - R^2$, so $r^{YY} = 1/(1 - R^2)$. By Theorem 1.6.3(b), $\det \mathbf{R} = \det \mathbf{R}_{XX} (1 - R^2)$.

Exercise B2.

Show that R^2 is unchanged if Y is replaced by $aY + b$ with $a \neq 0$, or the regressors by $\mathbf{B}^\top \mathbf{X} + \mathbf{c}$ with $\mathbf{B}$ nonsingular. Show that $\rho_{Y \cdot X}^2$ has the same invariance.

Exercise B3.

Show that $R^2 \rightarrow \rho_{Y \cdot X}^2$ with probability one as $n \rightarrow \infty$, without assuming normality, pro-

vided the rows are independent and identically distributed with finite second moments and Σ_{XX} positive definite.

14.3.4.3 C. Going deeper

Exercise C1.

With normal rows, show that the likelihood ratio statistic for $\rho_{Y.X}^2 = 0$ against an unrestricted alternative is $(1 - R^2)^{n/2}$, so the likelihood ratio test is the overall F test. *Hint:* under the null the covariance matrix is block diagonal; use the proof of [Proposition 14.3.1](#) and [Exercise \(B1\)](#).

Solution. With $\mathbf{S}$ the covariance matrix of the rows with divisor n , the maximized log-likelihood of a $(k + 1)$ -variate normal sample is $-\frac{n}{2}[\log \det \mathbf{S} + (k + 1)(1 + \log 2\pi)]$, by the argument in the proof of [Proposition 14.3.1](#) applied to all $k + 1$ variables. Under the null, Y and $\mathbf{X}$ are independent, the likelihood factorizes, and the maximum is the same expression with $\log(s_Y^2 \det \mathbf{S}_{XX})$ in place of $\log \det \mathbf{S}$. The likelihood ratio is therefore $(\det \mathbf{S} / (s_Y^2 \det \mathbf{S}_{XX}))^{n/2} = (1 - R^2)^{n/2}$, by Theorem 1.6.3(b) as in [Exercise \(B1\)](#). It is a decreasing function of R^2 , hence of F .